

NetSI PhD Bootcamp 2026 · Thursday, Sep 3 · 10am–12pm

Networks on the Cluster

Run your network science on Explorer

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How this session works

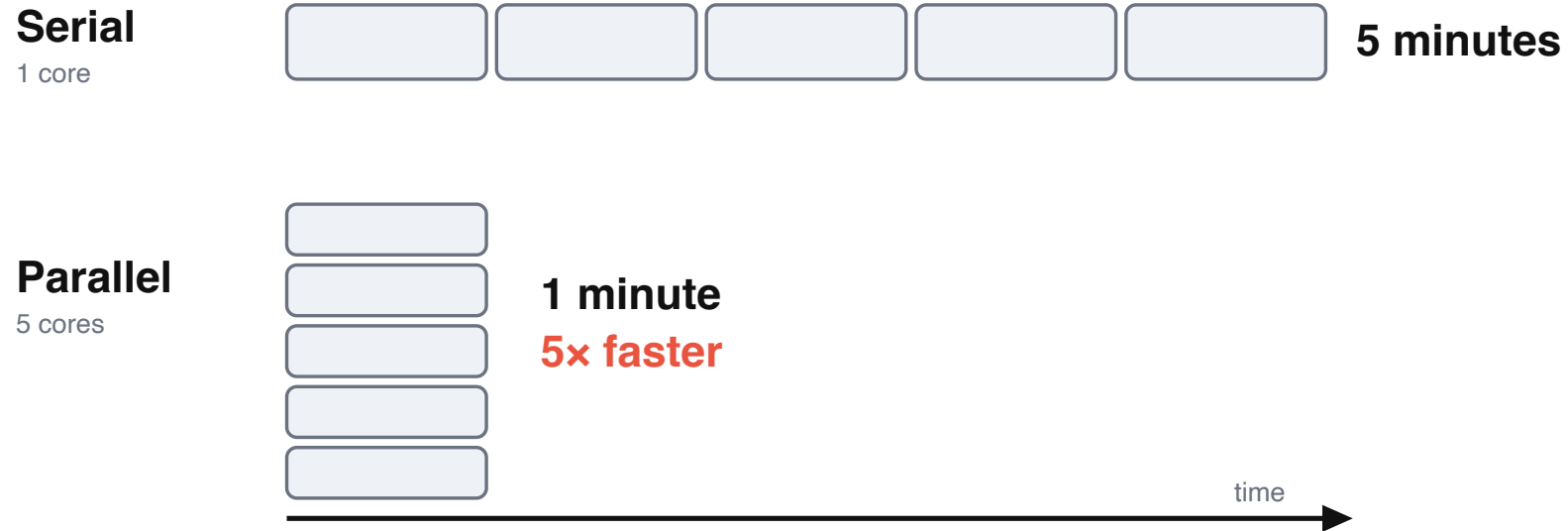
- ~45 min, guided speed-run: get onto Explorer and run a real job
- ~60 min, **Hackathon** pick one of three tracks
- ~15 min, lightning share-outs

By the end you'll have **submitted a parallel job on Explorer.**

Part 1: Speed-run

Why a cluster?

Parallel computing saves time by doing many things at once



Assuming the calculations don't depend on each other.

Why not one big workstation?

A single 64-core box hits walls:

- **Scalability**, what if you need 1,000 cores or 2 TB of RAM?
- **Availability**, a 2-week run holds the whole machine hostage
- **Efficiency**, expensive hardware sits idle between big runs
- **Concurrency**, 10 people can't all run at once

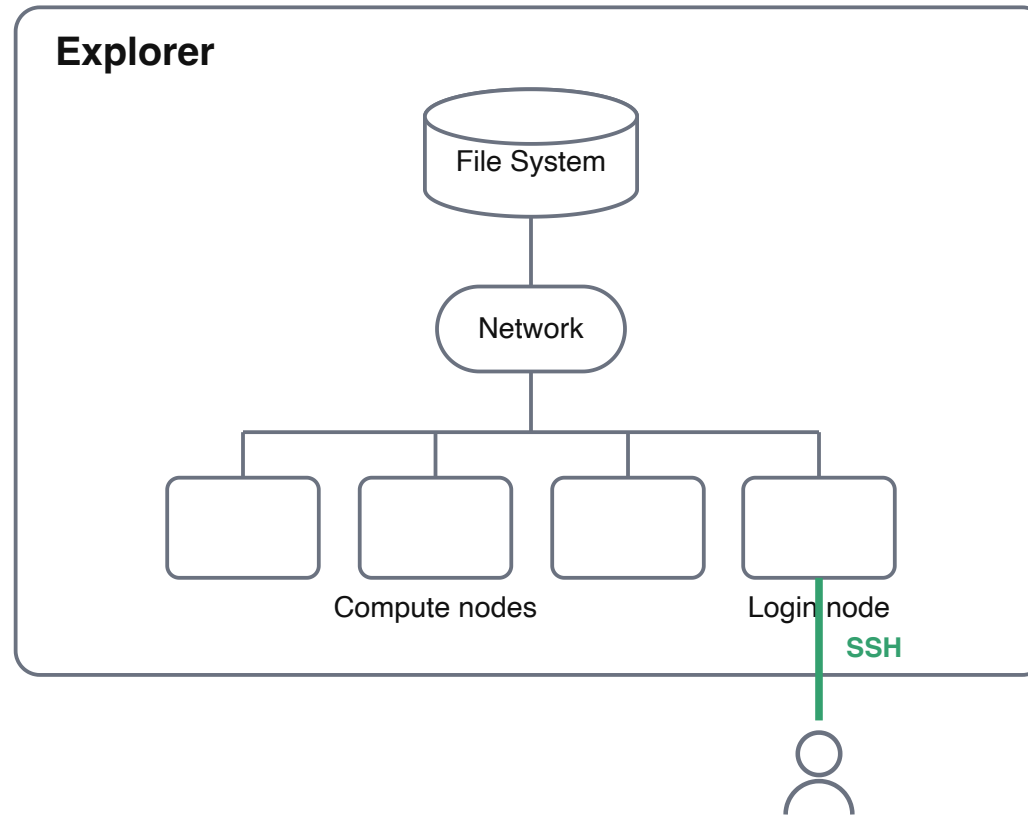
The solution: an HPC cluster

Many computers on a fast network, acting as one system

1,024 CPU nodes **~50k** cores **200+** GPUs

Explorer is Northeastern's cluster, hosted at MGHPCC in Holyoke, MA.

The pieces



Key terms

- **Node**, one computer in the cluster
 - **Login node**, where you edit files & submit jobs (**don't** compute here)
 - **Compute node**, the workhorses where jobs actually run
- **Partition**, a named pool of nodes, e.g. `short` (CPU), `gpu-short` (GPU)
- **Job**, a resource request + the commands to run
- **Job array**, many similar jobs from one script (`--array=1-N`)

Slurm runs the show



Simple Linux Utility for Resource Management

- Manages the **queue** of everyone's jobs
- Assigns your job to free compute nodes
- Keeps sharing **fair and efficient**

You describe **what you need**; Slurm decides **where and when**.

Slurm commands you'll use

```
sbatch job.sh           # submit a job
squeue --me             # your jobs    (ST: PD = pending, R = running)
scancel 12345           # cancel a job by id
srun -p short --pty bash -i  # interactive shell on a node
```

Are you ready?

Three checks before we get hands-on

1. **Connect** ▶ `ssh explorer` logs you in without a password
2. **Get the code** ▶ `git clone <repo>` (URL from instructors)
3. **Edit on the cluster** ▶ VS Code Remote-SSH or an OOD terminal

Not set up yet? No problem, we walk through SSH next (it's also in `SETUP.md`). Blocked? Use Open OnDemand in a browser, or pair with a neighbor.

Getting on Explorer

One-time setup, then it just works

Set up SSH once

Hands-on Session

Create a key (skip if you already have one):

```
ssh-keygen -t rsa      # press Enter through the prompts, no passphrase
```

Add a host alias, put this in `~/.ssh/config`:

```
Host explorer
  HostName login.explorer.northeastern.edu
  User YOUR_USERNAME      # your NU username, e.g. lastname.f
  IdentityFile ~/.ssh/id_rsa
```

Enable passwordless login

Hands-on Session

Copy your key up to Explorer (type your NU password once):

```
ssh-copy-id -i ~/.ssh/id_rsa.pub YOUR_USERNAME@login.explorer.northeastern.edu
```

Now this logs you straight in, no password, no typing the long host:

```
ssh explorer
```

Passwordless SSH is required for OOD / GUI apps to launch cleanly.

On Windows: same idea, two tweaks

Built-in OpenSSH works. Your config lives at `C:\Users\<you>\.ssh\config`

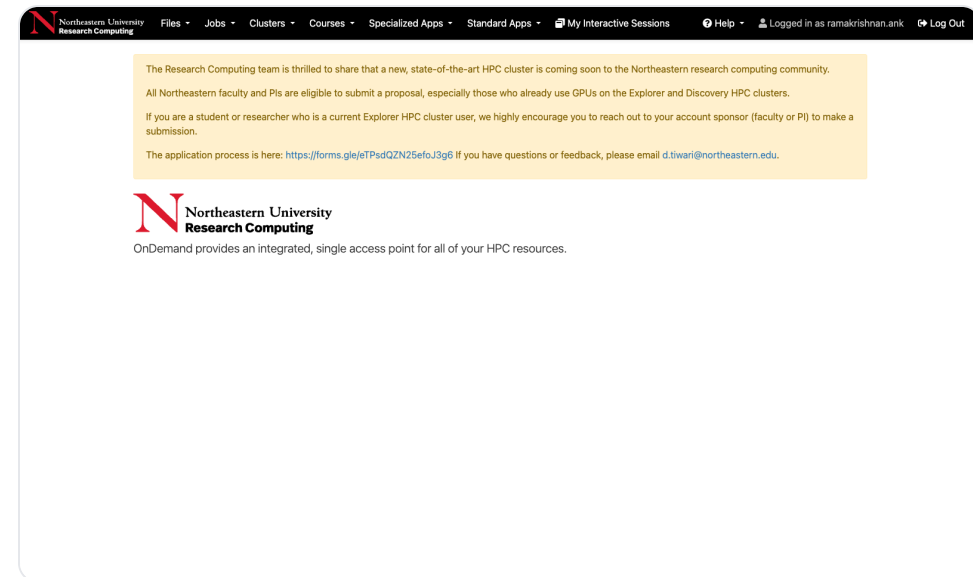
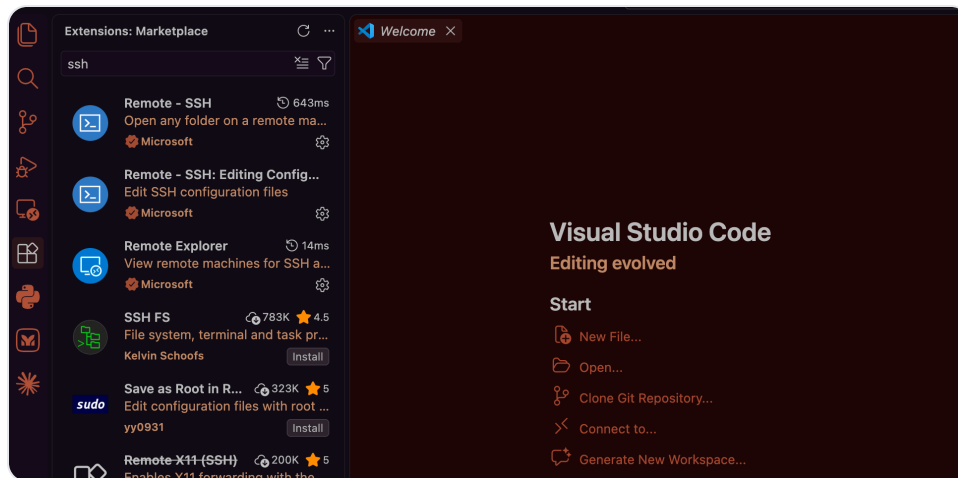
- **Easiest: use Git Bash.** Every command on the last two slides works as-is, **including** `ssh-copy-id`.
- **PowerShell** works too, but has no `ssh-copy-id`. After `ssh-keygen` and the same config, copy your key up:

```
$k = "$env:USERPROFILE\.ssh\id_rsa.pub"  
type $k | ssh explorer "mkdir -p ~/.ssh; cat >> ~/.ssh/authorized_keys"
```

VS Code Remote-SSH reads the same `config`, so `explorer` shows up the same way.

Two doors into Explorer

- **VS Code Remote-SSH** ▶ command palette ▶ **Connect to Host** ▶ pick **explorer**
- **Open OnDemand** ▶ web portal at ood.explorer.northeastern.edu



VS Code Remote-SSH gives you an editor and terminal on the cluster (left). Open OnDemand runs files, JupyterLab, and job monitoring in the browser (right).

Shell survival kit

```
pwd           # where am I?  
ls            # what's here?  
cd my_project # move into a folder  
mkdir bootcamp # make a folder  
cat notes.txt # print a file
```

Everything below runs in the terminal, VS Code's or OOD's, same thing.

Jump onto a compute node

Hands-on Session

The login node is shared. For a quick test, grab your own compute node:

```
hostname                # login node, e.g. explorer-01

srun -p short --time=00:30:00 --pty bash  # ask Slurm for a live shell
# (waits a moment while it allocates)

hostname                # now a compute node, e.g. c0613
exit                    # release it, back to the login node
```

Great for quick tests and debugging. For real work, submit a **batch** job (next).

Get the example code

Hands-on Session

```
# clone this bootcamp's repo (URL from your instructors)  
git clone https://github.com/TheCheshireCat/bootcamp-2026  
cd bootcamp-2026/hackathon/cluster  
ls  
# 00-hello  track-a-parallel-sim  track-b-gnn-gpu  track-c-big-graph
```

Everything we run today lives here.

Software: environment modules

Major scientific software is pre-installed, just load it

```
module avail          # browse everything available  
module load anaconda3 # unlock the `conda` command  
module list           # what's loaded now
```

Software: conda environments

Your own isolated, reproducible Python

```
module load anaconda3
conda create -n bootcamp python=3.11 networkx numpy pandas matplotlib
source activate bootcamp

which python3           # now points inside your env
# /home/YOUR_USERNAME/.conda/envs/bootcamp/bin/python3
```

Start this now: the array-job walkthrough coming up (Steps 1-2) runs in `bootcamp`. Solving takes a few minutes, so kick it off, then keep following along. (The hello job itself just uses `base`.)

Anatomy of a batch file

```
#!/bin/bash
#SBATCH --job-name=hello-cluster
#SBATCH --partition=short          # the old "express" is retired
#SBATCH --time=0-00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1

module load anaconda3
source activate base              # hello_cluster.py has no dependencies
python3 hello_cluster.py
```

Resources up top (`#SBATCH`), then the **commands to run**.

Submit, watch, collect

Hands-on Session

```
cd 00-hello
sbatch job.sh           # ▶ Submitted batch job 49725148

watch -n 1 squeue --me  # ST: PD=pending, R=running (Ctrl-C to quit)

ls
# hello_cluster.py  job.sh  slurm-49725148.out
```

Open the `slurm-*.out` file, that's your job's output.

One script, many runs

The real superpower: job arrays

ER graphs, how does average degree depend on connection prob p ?

- 11 values of p $\times$ 100 trials = **1,100 simulations**, all independent
- One array fans them out; a second job aggregates + plots

```
#SBATCH --array=1-11      # one task per p-value
```

Result: $\bar{k} = p(n - 1)$. Let's actually run it.

The array job: two steps

One folder, two jobs: simulate, then plot

```
cd track-a-parallel-sim
ls
# sim.py job_sim.sh plot.py job_plot.sh
```

- `sim.py` + `job_sim.sh`: run **11 tasks** (one per `p`) ▶ `results/p_*.csv`
- `plot.py` + `job_plot.sh`: read the CSVs ▶ `figs/avg_degree_vs_p.png`

Step 1: run the simulation

Hands-on Session

```
# open job_sim.sh, set your email (or delete the --mail lines)

sbatch job_sim.sh          # ▶ Submitted batch job 51234567
watch -n 1 squeue --me     # 11 array tasks appear, one per p

ls results/
# p_0.0.csv p_0.1.csv ... p_1.0.csv
```

Each task wrote its **own** CSV. That's the array at work.

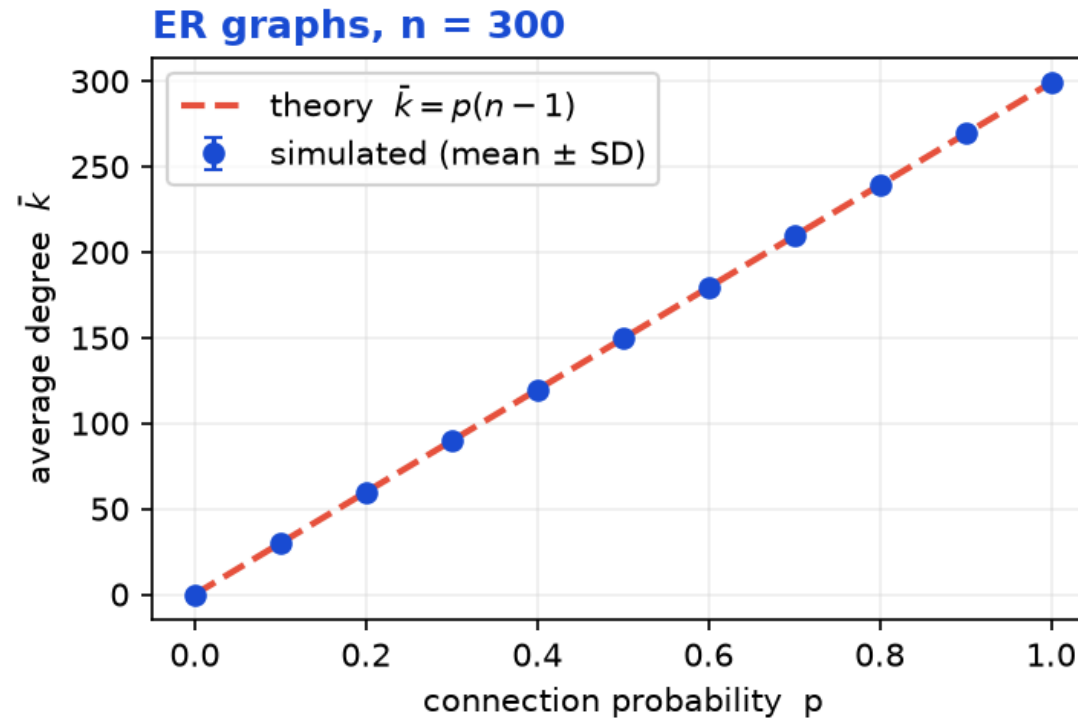
Step 2: plot the results

Hands-on Session

```
sbatch job_plot.sh  
ls figs/  
# avg_degree_vs_p.png
```

- Open `figs/avg_degree_vs_p.png` , that's your result ▶

The payoff



11 array tasks, one figure: the simulated points land right on the theory line $\bar{k} = p(n-1)$.

You just ran a real parallel study. Now go make it **your own**.

If you get stuck

Symptom	Fix
<code>ssh: Permission denied</code>	rerun <code>ssh-copy-id</code> , or use OOD instead
Job sits in PD (pending)	queue is busy, lower <code>--time</code> / <code>--mem</code> or wait
<code>conda: command not found</code>	run <code>module load anaconda3</code> first
CUDA: <code>no kernel image</code>	pin <code>--gres=gpu:v100-sxm2:1</code> + matching <code>cuda/</code>

Still stuck? Pair with a neighbor or grab an instructor, don't burn 20 minutes solo.

Part 2: Hackathon

Pick a track. Explore Explorer.

Choose your own adventure

Track	Difficulty	Compute
A · Parallel simulations	BEGINNER	CPU (short)
B · GNN on GPU	INTERMEDIATE	gpu-short
C · Big-graph generate & analyze	INTERMEDIATE	CPU (short)

Starter code + step-by-step: [hackathon/cluster/](#)

Track A: Parallel simulations

BEGINNER

Take the ER job array and make it **your** experiment

- Sweep a parameter, run trials in parallel, aggregate into one figure
 - Learn: `--array` , per-task output files, a simulate ▶ plot workflow
 - Everything runs on `short` , no GPU needed
- ▶ `hackathon/cluster/track-a-parallel-sim/`

Track A: what could you study?

Same pattern every time: **sweep a parameter, measure a property, plot.**

- **Giant component:** sweep mean degree in ER, watch it jump at `mean degree = 1`
- **Small world:** sweep rewiring in Watts-Strogatz, watch path length collapse
- **Scale-free:** sweep `m` in Barabási-Albert, look at the degree tail
- **Robustness:** remove nodes randomly vs by degree, track the largest component
- **Epidemics:** sweep infection rate, find where outbreaks take off

Full menu with hints: `track-a-parallel-sim/README.md`

Track B: GNN on GPU

INTERMEDIATE

Train a graph neural net on a real GPU

- Node classification (Cora / karate club) with PyTorch Geometric
 - Request a GPU: `-p gpu-short --gres=gpu:v100-sxm2:1`
 - Learn: GPU jobs, `nvidia-smi`, matching CUDA to your build, CPU fallback
- ▶ `hackathon/cluster/track-b-gnn-gpu/`

Picking a GPU (don't just say `gpu:1`)

Explorer mixes old and new cards, the type changes what runs:

<code>--gres=gpu:TYPE:1</code>	Card	Good for
<code>v100-sxm2</code> ★	V100, 32 GB	our default, always works, tons available
<code>a100</code>	A100, 40–80 GB	bf16 / TF32 / big memory
<code>h200</code>	H200, 141 GB	newest; needs recent CUDA + matching PyTorch
<code>t4</code>	T4, 16 GB	small / inference only



Mismatched CUDA ▶ **"no kernel image is available for execution on the device."**

Pin `v100-sxm2` and you'll avoid it.

Track C: Big-graph generate & analyze

INTERMEDIATE

Push the cluster with large networks

- Generate big graphs across an array, measure properties, aggregate
 - Learn: scaling with array jobs, memory requests (`--mem`), combining results
 - Watch a task run out of memory, then fix the request
- ▶ `hackathon/cluster/track-c-big-graph/`

Share-outs

60 seconds each

- **What I ran** (which track, what question)
- **What worked**
- **What broke**, and how you dealt with it
- **What surprised you**

Best practices & resources

Good habits, and where to go next

Play nice on shared hardware

Explorer is shared by everyone, so be a good neighbor

- **Never** run heavy work on the login node, submit with `sbatch` or grab a node with `srun`
- Set a realistic `--time`, and cap array concurrency: `--array=1-100%10`
- Job stuck, or won't release its nodes? Email rchelp@northeastern.edu

Put files in the right place

Location	Use it for	Watch out
<code>~/</code> (home)	code, small files	small quota
<code>/scratch</code>	large temp data	purged periodically
<code>/projects/netsi</code>	lab / project storage	shared with NetSI

Don't keep anything you can't lose in `/scratch`.

Write code future-you can run

- **Docstrings + comments** that explain **why**, not what
- **Log, don't print**, so you can debug a job that failed at 3am
- Feed the Slurm task id in as a CLI arg, index into a parameter list
- With an LLM, be the **architect**: design the workflow, then let it fill in code

```
task_id = int(sys.argv[1])      # from #SBATCH --array
params  = PARAM_SETS[task_id]  # this task's parameters
```

Make it reproducible

- **Use git**, commit as you go, push to GitHub as an off-cluster backup
- **One conda env per project**, so package versions don't collide
- Future-you and your collaborators can rebuild the exact run

```
conda create -n myproject python=3.11 networkx numpy  
git init && git remote add origin <your-repo>    # backup + history
```

Where to get more compute

Develop on Explorer first, then scale out

- **AICR** rc.northeastern.edu/aicr: the Mass. AI Hub cluster, B200 + RTX6000 Pro GPUs, an extension of Explorer. Your **PI submits a project proposal**.
- **NAIRR pilot** nairrpilot.org: free national GPU allocations
- **Free notebooks**: Colab colab.research.google.com, Kaggle kaggle.com

Full list, plus free platforms and cheap rentals: **RESOURCES.md**

Keep learning

- Explorer docs: rc-docs.northeastern.edu
- **GitHub Student Pack** (free Copilot, credits): education.github.com/pack
- Learn git by doing: learngitbranching.js.org
- PHYS7332 network science book: asmithh.github.io/network-science-data-book

Everything above, with links, lives in **RESOURCES.md**.

Takeaways

- You can now: **connect, build an env, submit jobs, fan out with arrays**
- The pattern behind almost all of it: **many independent jobs, one script**
- Docs: rc-docs.northeastern.edu

Same problems, but faster, bring this to the **2pm Agentic session**.